DISCOVERATHON 2026 · CHALLENGE 2

Predicting hERG Channel Blockade

A model built for trust, not just accuracy

Presented by

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Trust starts with the data

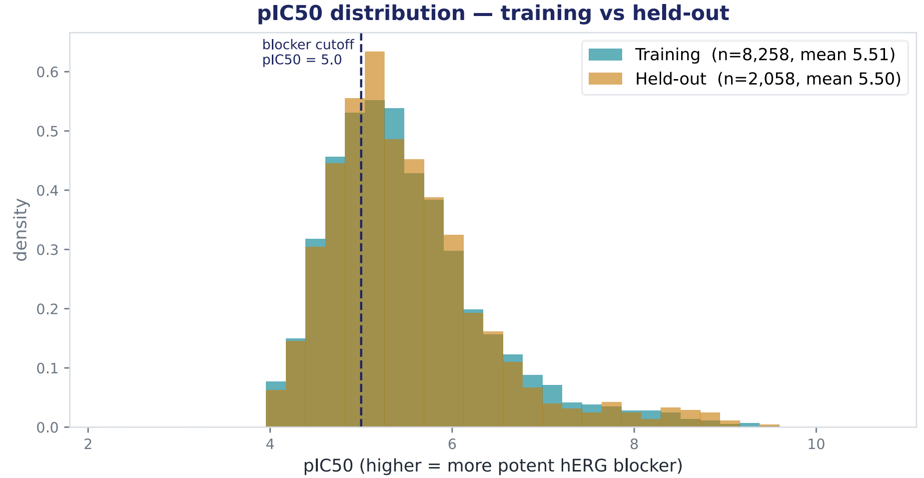
What it is. All **16,650 unique molecules**; each a SMILES structure with a yes/no hERG-blocker label, balanced at **48.8% blockers** are used for the classification task, and the **10,316** of them that carry an exact measured pIC50 value are used for the regression task.

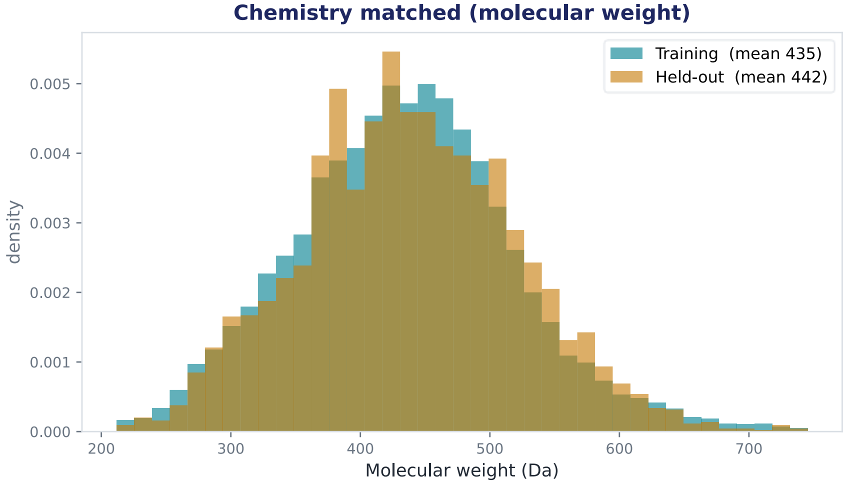
Inclusion criteria. **ChEMBL 37 (CHEMBL240) + BindingDB**, one gated extraction:

- ❑ confidence ≥ 7.
- ❑ human target,
- ❑ wild-type only (mutant assays removed).
- ❑ validity/duplicate cleared.
- ❑ Blocker = pIC50 ≥ 5.0 (10 μM).
- ❑ RDKit-standardized.
- ❑ InChIKey-deduplicated.

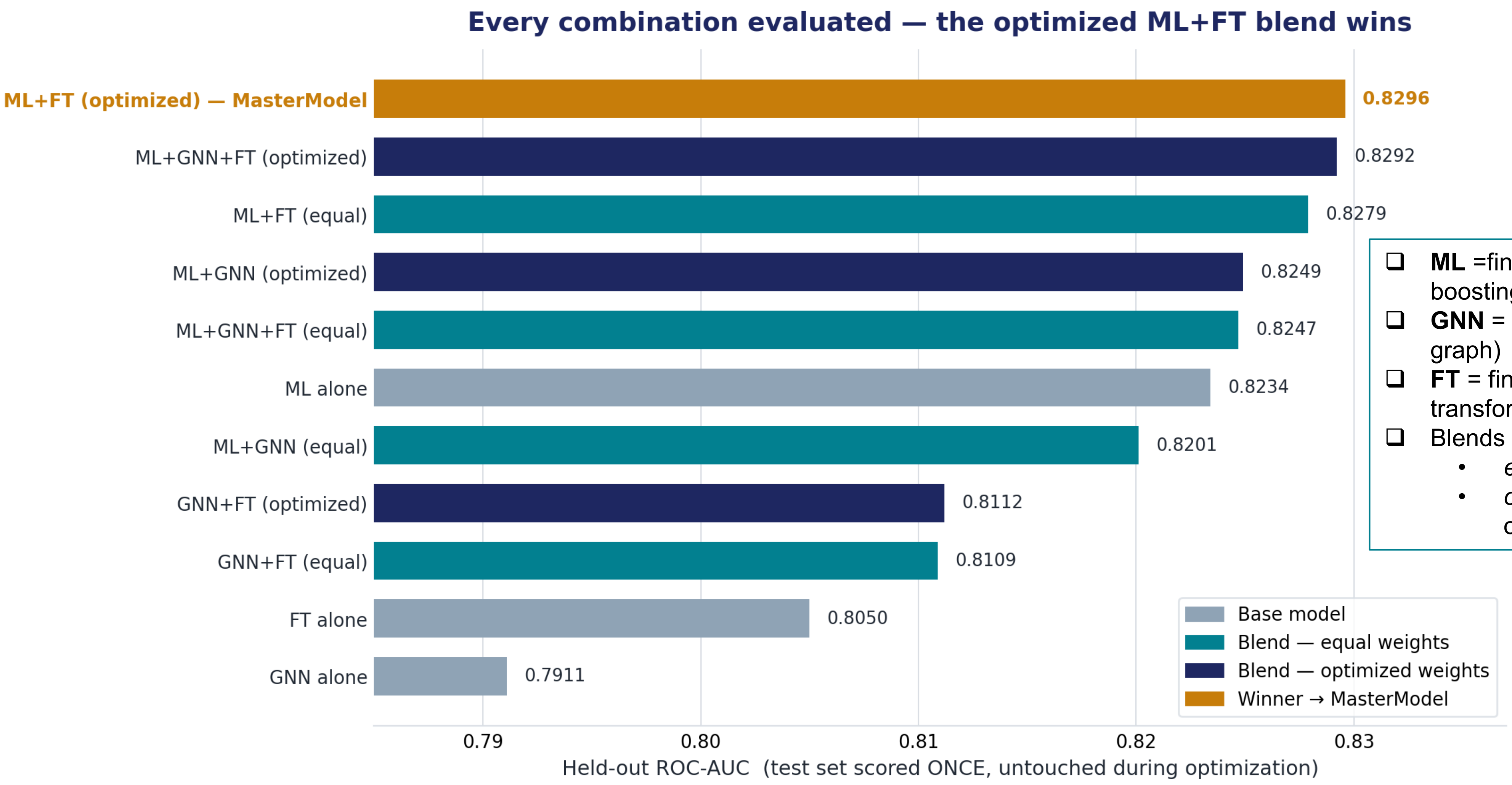
Split & held-out. 20% held out once, up front, by Bemis–Murcko scaffold (seed 42); **zero scaffold overlap with training**, for generalization of the model. Class balance preserved. It is never seen during training or model selection.

Why the external. **957 marketed drugs** with clinical cardiotoxicity labels (DICTrank + CredibleMeds); testing whether a hERG-only model can predict real cardiotoxicity, which extends beyond hERG. Validation only.





Base models, blends, and the winner (all scored on the untouched held-out set)

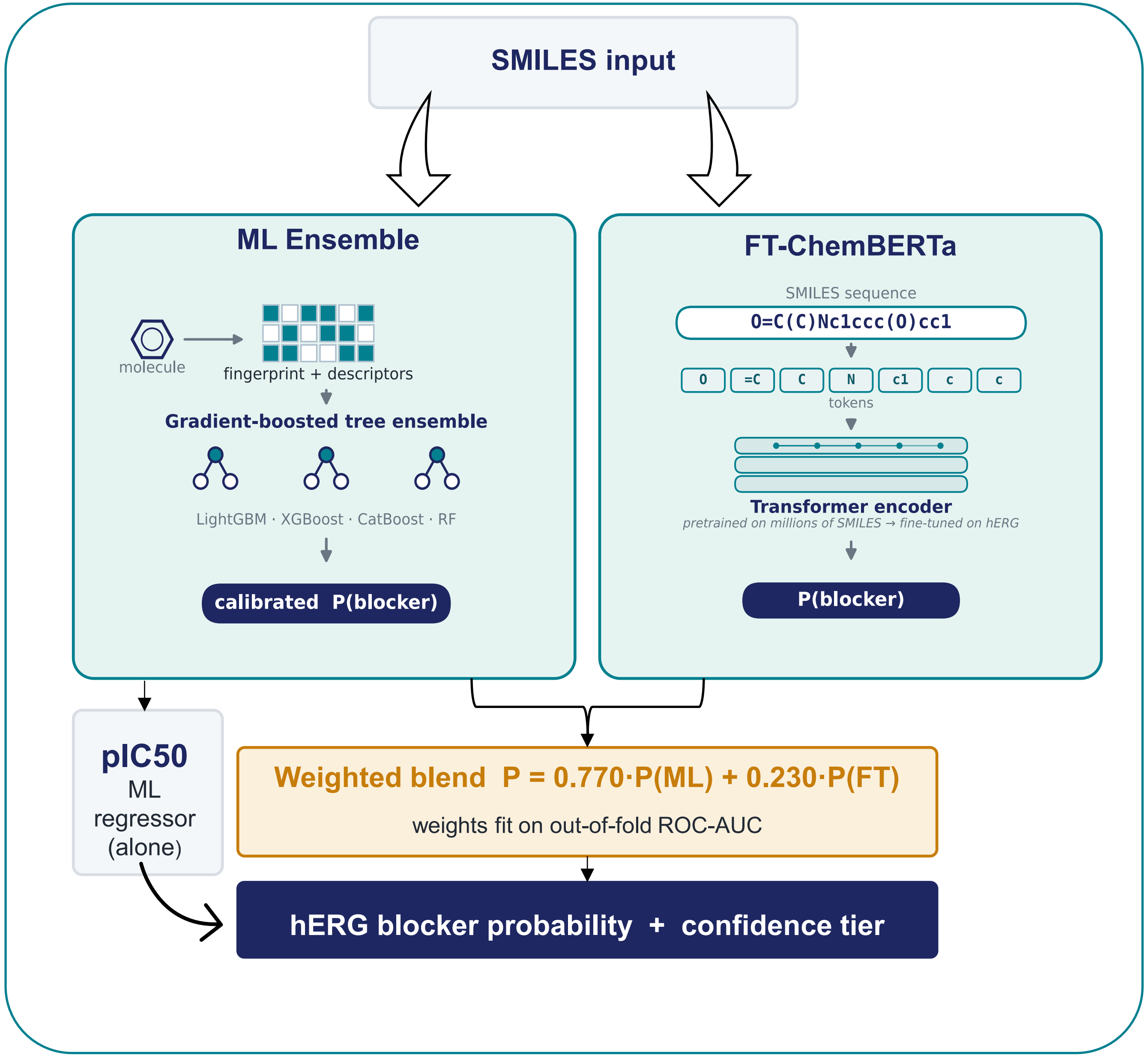


- **ML** =fingerprint/descriptor gradient-boosting ensemble ·
- **GNN** = graph neural network (molecular graph)
- **FT** = fine-tuned ChemBERTa transformer (SMILES sequence).
- Blends combine their probabilities:
  - *equal* = averaged,
  - *optimized* = weights tuned on out-of-fold data.

Trust move: the test set was scored only ONCE, after weights were fixed on out-of-fold data : no tuning on the number we report.

ARCHITECTURE & BLEND LOGIC

Two decorrelated models, combined by evidence



ML Ensemble

How it reads a molecule. As a numeric feature table; Morgan/ECFP fingerprints plus RDKit physicochemical descriptors, capturing the molecule's substructures and properties.

What it is. A calibrated stack of gradient-boosted trees (LightGBM, XGBoost, CatBoost, Random Forest, Extra Trees). It's the strongest single model (held-out ROC-AUC ≈ 0.82) and also serves as the sole pIC50 regressor.

FT-ChemBERTa

How it reads a molecule. As a SMILES sequence; a transformer processes the molecule token-by-token, the way a language model reads text.

What it is. ChemBERTa pretrained on millions of unlabeled SMILES, then fine-tuned end-to-end on our hERG data. Fine-tuning lifts the base model from ~0.77 to ~0.80.

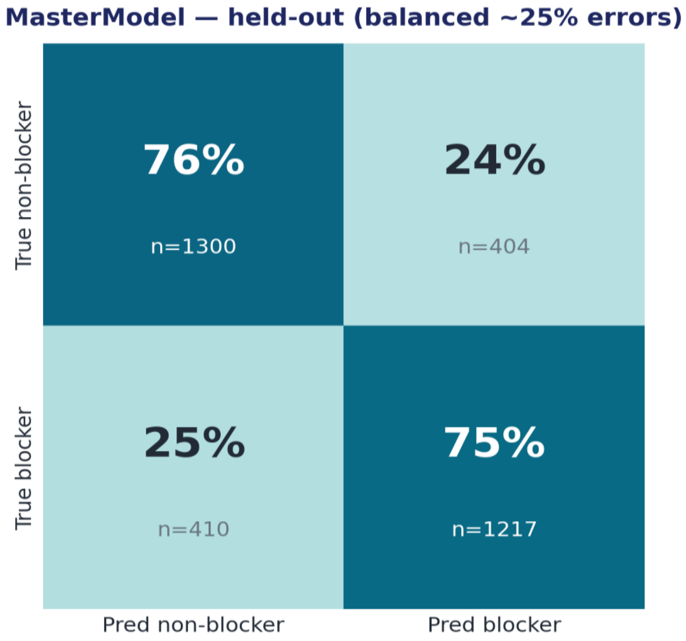
Why blend

Decorrelated errors. A fingerprint/tree model and a fine-tuned transformer 'see' molecules differently and make different mistakes, bending them cancels error and cuts variance.

Leakage-safe weights. The 0.77/0.23 weights were optimized on out-of-fold predictions, never on the held-out test set, so the reported number isn't tuned.

RESULTS — HELD-OUT

The full scorecard



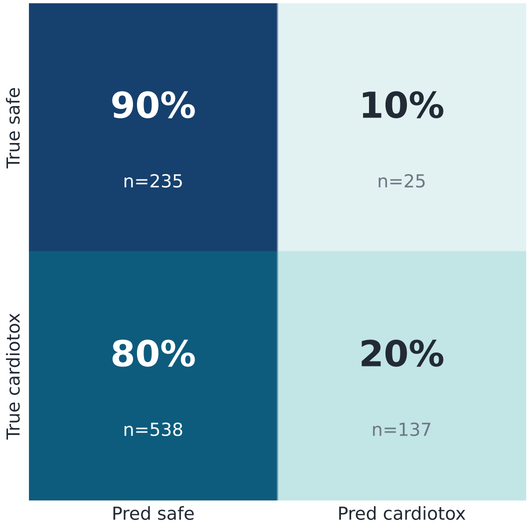
Held-out ROC-AUC ≈ 0.83, balanced ~25% error either way : a well-calibrated classifier on its trained task.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
ML Ensemble (baseline)	0.746	0.744	0.731	0.738	0.823	0.821
GNN (baseline)	0.722	0.702	0.749	0.725	0.791	0.780
ChemBERTa (baseline)	0.700	0.701	0.674	0.687	0.772	0.766
FT-ChemBERTa (baseline)	0.726	0.706	0.752	0.729	0.796	0.786
MasterModel (final)	**0.750**	**0.749**	**0.734**	**0.741**	**0.830**	**0.825**

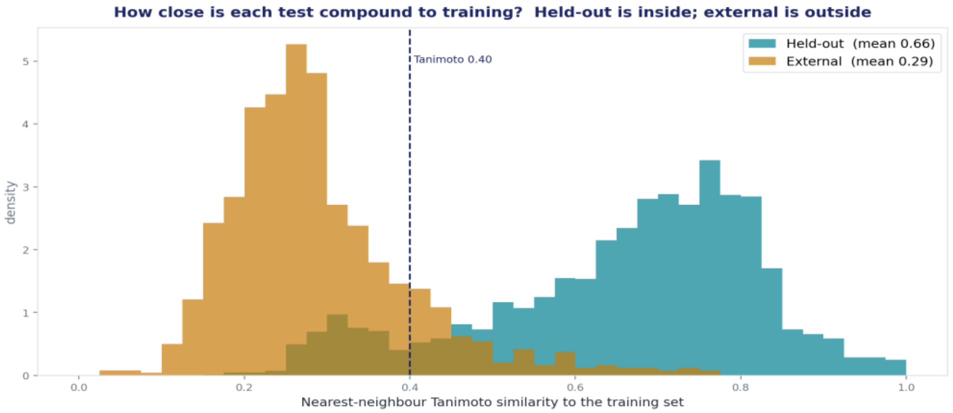
Trust in action: keeping only the model's HIGH-confidence predictions raises held-out ROC-AUC to 0.89 at 58% coverage; the model knows when to abstain.

Where it holds, and where it doesn't.

MasterModel — external (conservative: misses cardiotoxins)



Metric	Held-out (hERG)	External (cardiotox)
Accuracy	0.756	0.398
Precision	0.751	0.846
Recall	0.748	0.203
F1	0.749	0.327
ROC-AUC	**0.832**	**0.610**
PR-AUC	0.827	0.803



86% of external compounds fall below 0.40 similarity to any training molecule, vs only 11% of held-out — quantifying the distribution shift.

Held-out sits inside the training chemistry (mean similarity 0.66); the external drugs are a genuine distribution shift (0.29)

What these numbers mean in the real world

Rule-in, not rule-out. When it flags a drug, it's right ~85% of the time (actionable red flag) and it rarely mislabels safe drugs (90% specificity). But it catches only ~20% of cardiotoxic drugs, because most clinical cardiotoxicity is *non-hERG* (calcium/sodium, mitochondrial, structural), which a hERG model can't see. ROC-AUC 0.61 = real but partial signal.

A positive means *investigate*; a negative does *not* mean safe.

Advantages built in by design

Confidence on every prediction

Applicability-domain + boundary flags give a HIGH/MEDIUM/LOW tier. Validated: **HIGH-confidence = 0.89 ROC-AUC vs 0.71 for the rest,** the model knows which of its own answers to trust.

No leakage, no inflation

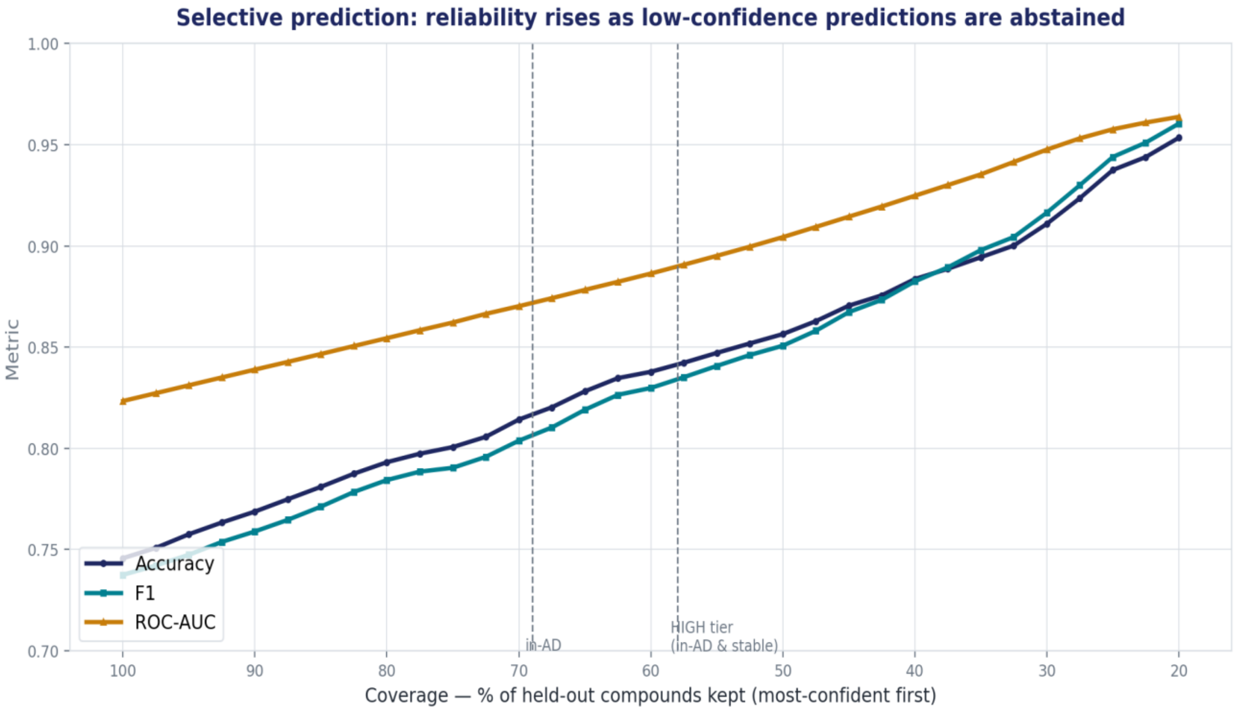
Weights and thresholds fixed off-test; the held-out set scored only once (never tuned on). De-duplicated by InChIKey, external held out the same way. No compound leaks across splits.

Generalizability

Scaffold split (zero overlap) tests new chemistry, not memorized analogues. Random splits leak look-alikes and inflate scores. Reinforced by a pretrained + fine-tuned transformer (best out-of-distribution) and a decorrelated blend.

Reproducible & robust

master_predict.py regenerates submission.csv to 1e-4 from a clean clone; a decorrelated blend is steadier than any single model.



Full coverage (100%): Acc 0.75 / AUC 0.82. At 58% coverage (HIGH-confidence tier): Acc 0.82 / AUC 0.88 — accuracy bought by abstaining on the hard 42%.

Reliability rises as the least-confident predictions are abstained

LIMITATIONS

What we do not claim

hERG ≠ all cardiotoxicity

hERG isn't the whole picture. Cardiotoxicity has many causes; our model sees only the hERG one, so it can never catch the drugs that harm the heart by other routes, no matter how we adjust it.

Context-blind

Molecule only, not the patient. It predicts whether a drug *can* block hERG, not whether it will actually cause harm, which also depends on the dose, how much reaches the heart, and other channels it hits. (Verapamil blocks hERG yet is cardiac-safe.)

Not a hard filter

Not a pass/fail gate. "Safe" ≠ safe , use it to flag suspicious compounds, not to clear them. Turn its caution up and it catches ~2.4× more risky drugs (worth the extra false alarms in a safety screen).

External labels are a proxy

A rough yardstick, not ground truth. The clinical labels are expert verdicts per drug (not patient data), ignore dose/exposure, and were partly based on hERG evidence already, so we treat them as a sanity check on direction, not proof of correctness.

What we built, and what we declare

A trustworthy hERG predictor

strong on its trained task (held-out ROC-AUC ≈ 0.83), honestly bounded on clinical cardiotoxicity (~0.61), with a confidence tier on every prediction and a fully reproducible pipeline. We prioritized reliability and honesty over a higher headline number ; deploy it as an early hERG-liability filter and one input to a broader cardiac-risk assessment.

Integrity declaration

Data: ChEMBL 37 + BindingDB (hERG bioactivity); FDA DICTrank + CredibleMeds QTdrugs (external, validation-only). Sources & versions documented; external set is a curated proxy, not raw clinical data.

Tools: RDKit, scikit-learn, XGBoost/LightGBM/CatBoost, PyTorch + HuggingFace Transformers (ChemBERTa).

AI assistance: AI coding assistants were used for implementation; the scientific reasoning, model choices, and analysis are the team's own.

Assumptions & risk: external labels are drug-level, exposure-agnostic, and partially circular with hERG; all claims are bounded to hERG-mediated cardiotoxicity.

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***THANK YOU ***